Video Crowd Detection and Behavior Analysis

Antoine Fagette
Thales
Sorbonne Universités, Université Pierre et Marie Curie
Image and Pervasive Access Laboratory, UMI CNRS 2955

Thesis supervised by Prof. Daniel Racoceanu (Sorbonne Universités, UPMC-Univ.Paris 6)
and Dr. Jean-Yves Dufour (Thales Services, Palaiseau)

1 Problematic

Crowd monitoring is currently one of the most challenging and impactful topics within the computer vision field dedicated to video-surveillance. Understanding, through video analysis, the behavior of a crowd, is a real need that the professionals of the domain express even more urgently, since CCTV networks are now present and growing faster than ever in every public places. Moreover, this phenomenon is often happening without any consideration for the human resources required for the monitoring, leading to a misuse of the gathered data.

Crowd behavior analysis can either be used to manage the environment (*e.g.*, to adapt it, in order to enhance the circulation conditions of the pedestrians) or to detect abnormal events (*e.g.*, stampedes, terrorists attacks, people fainting, etc.) that may lead to hazardous situations. In the first case, simulations can be run in order to identify the best answers to the most likely events before they happen. However, this solution does not allow an adaptive response to every possible evolution and may lead to counter-efficient organization of the space, due to unattended events.

The challenge of crowd behavior analysis through computer vision, is to enable a reaction as soon as an event arises with a correct interpretation, based on the videos that are captured by the cameras of the CCTV network, and deployed over the situation to be handled.

In order to fulfill this task, several studies have already been leaded, some of them with the idea to build a physical model of crowd. One of the most popular models is the Social Force model elaborated by Helbing and Molnár [4] using a microscopic approach for crowd representation and simulation. Such approach consists in modeling the crowd as a set of individuals that interact together on the basis of local social forces. Conversely, some authors also consider a crowd as a whole fluid entity. This has led researchers to the path of the fluid mechanics with the main assumption that a crowd can be driven using a macroscopic approach.

2 Thesis Outline

This thesis concentrates its research efforts on the similarity between a fluid and a crowd. It exploits this resemblance and follows a holistic approach of the crowd, motivated by the idea that the individual can sometimes be invisible in a crowd, even though he contributes to its global behavior.

Preliminary studies, focusing on the analysis of the raw optical flow, have led us to compute two versions of the "temperature" of the crowd: the first is taken as is from Helbing's model

[3] and the second is directly derived from the gas kinetic theory. The results, displayed on Figure 1, are promising for the detection specific events and show that the temperature is a good feature for some behaviors, such as inconsistency or sudden changes in the flow motion.

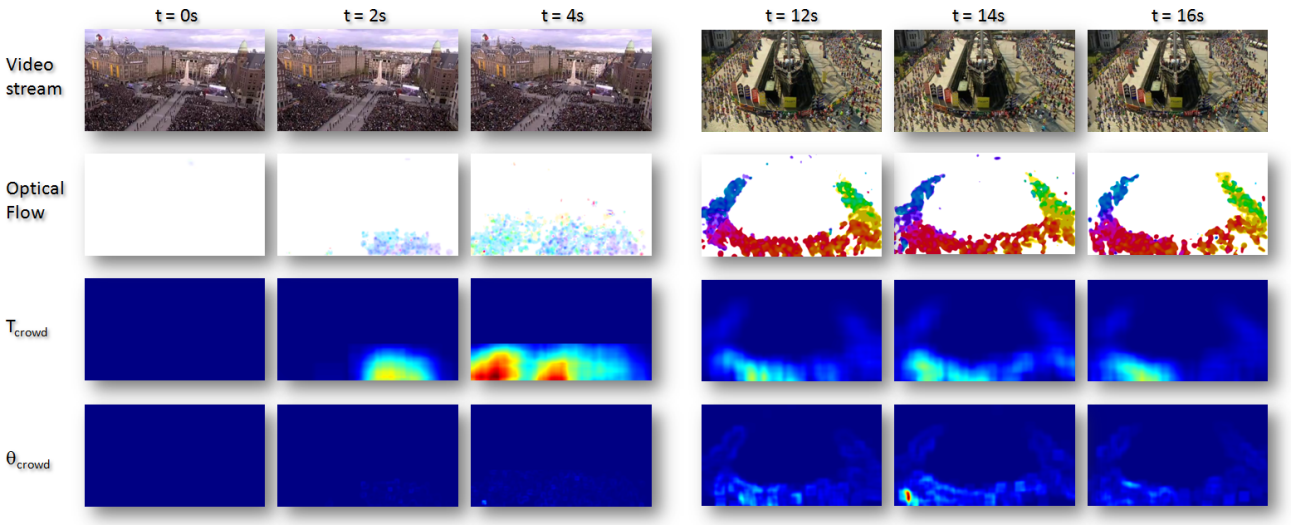


Figure 1: Results of the preliminary study for the analysis of the optical flow. T_{crowd} represents the temperature derived from the gas kinetic theory, θ_{crowd} represents the temperature as defined by Helbing.

However, the study of the raw optical flow presents also some drawbacks. Being dense, optical flow allows an evaluation of the displacement of every pixels between two frames. Nevertheless, this approach is a very short term one, and one cannot perform any tracking by simply using it. Studying the crowd's behavior requires more than the study of its evolution between two frames. Hence, we need an algorithm, capable to track the flow of pedestrians. The first approach for crowd flow tracking consists in segmenting and tracking each individuals. This works for image with a good enough resolution, or rather sparse crowds. Our holistic approach is complementary, as it aims at performing the same task for crowds too dense and/or images with a too low resolution to achieve good performances on pedestrian's detection.

With our idea to explore the analogy between a crowd and a flow, the Particle Video algorithm described by Peter Sand [5] fits our needs. The particles, just like fluid particles, are driven by the (optical) flow. The whole study has therefore been oriented on the adaptation of the particle video algorithm for crowd tracking and analysis. This interrogation ended up with the description of the system illustrated on Figure 2.

The proposed system is composed of 5 main blocks. The first and the second block allow evaluating the two basic physical features parameters of the crowd: speed and density. The "Estimation of the local long-term motions" block is where the particle video algorithm is run. The cloud of particles must be statistically representative enough of the crowd in order to assume that their measured trajectories can reflect the crowd dynamic behavior. The "Estimation of the global long-term motions" block is where the features, such as the temperature, are computed. These features are no longer computed directly on the optical flow but derived from the particles observed motion. Finally, to come up with a finer analysis of the behavior, the system could refer to a knowledge base or perform some learning itself. Of course, for this approach to be usable on a CCTV network, we should address the problematic of the multi-camera usage. Besides, in order to deal with Smart Environments and Smart Cities, a multi-modality compliance need to be considered.

This whole process is leading to the evaluation of the crowd parameters, the estimation of the crowd flow motions and the computation of behavior features, by ending up with the

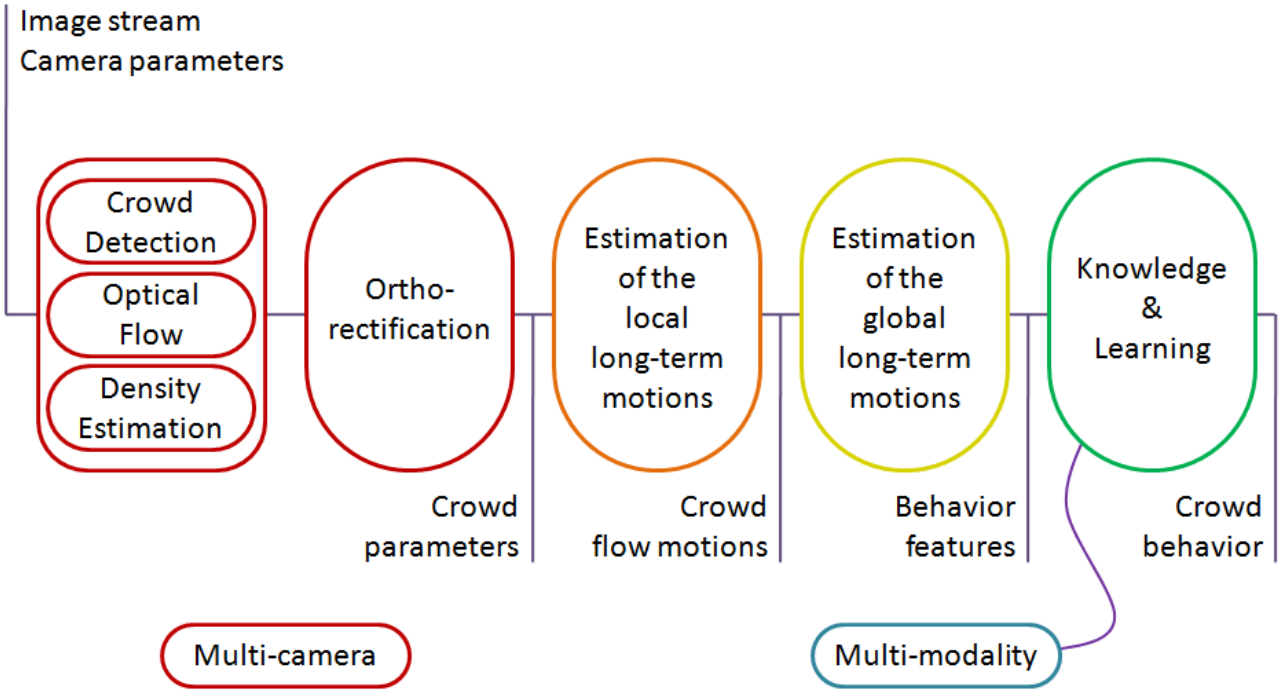


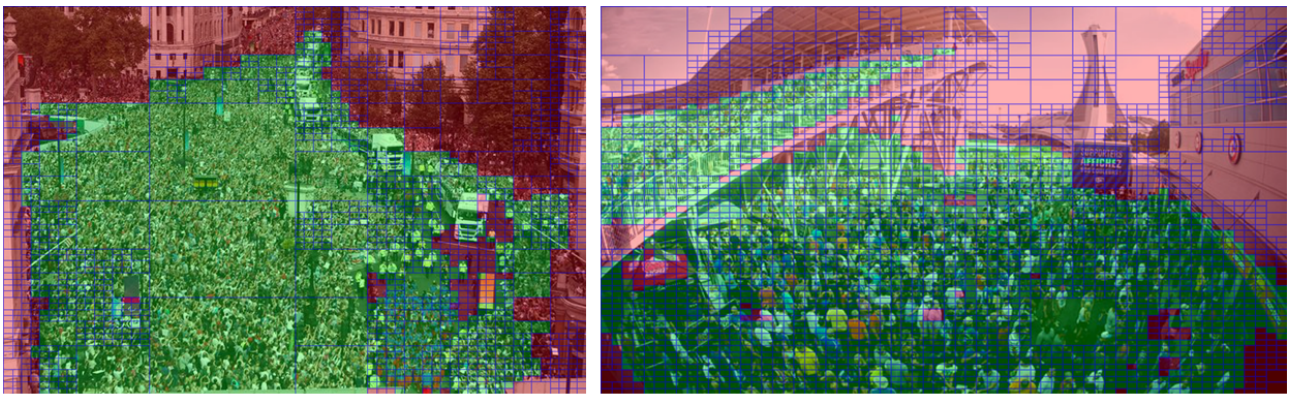
Figure 2: Framework, consequence of the preliminary study.

analysis of the crowd behavior.

3 Thesis Outcome

Based on the framework described on Figure 2, this thesis has addressed three main problems: the detection of the crowd, the estimation of its density and the tracking of the flow.

The contribution to crowd detection introduces a totally unsupervised method for the detection and location of dense crowds in images without context-awareness. After retrieving multi-scale texture-related feature vectors from the image, a binary classification is conducted to identify the crowd and the background. The algorithm can be operated on images without any prior knowledge of any kind, being totally unsupervised. The Figure 3 shows some results produced by our algorithm.



(a) (b)

Figure 3: Crowd detection

So far, all the density estimation algorithms that have been designed for crowd density estimation, were giving a discrete evaluation of the density (low, medium, high). For the density estimation, we are developing an original algorithm using supervised learning to give a numerical estimation of it.

The two previous contributions had for only goal to enable the adaptation of the particle video algorithm to crowd analysis with our constraints. The adaptation that we are doing is at two levels: the setup of the particles on the image and their advection process. As opposed to most of the existing implementation of particle video for crowd analysis, we are not setting our particles on the image on a regular grid. We rather spread our cloud of particles on the image, according to the estimated crowd density. This cloud of particles may be thus considered as statistically representative of the crowd. Therefore, each particle has physical properties that enable us to assess the validity of its behavior according to the one expected from a pedestrian, and to optimize its motion guided by the optical flow. This leads us to three applications: the detection of the entry and exit areas of the crowd in the image, the detection of dynamic occlusions and the possibility to link entry areas with exit ones, according to the flow of the pedestrians.

This thesis represents the first brick of a wider project initiated by Thales and IPAL on crowd management for safe/smart cities using video-surveillance and simulation. The blocks of the system that have been studied and implemented are the primary tools for a deeper analysis of crowds. Leveraging on the work that has been done, one can clearly foresee the future steps that are to be explored: find new behavior features, develop some learning methods to recognize effectively the crowd behaviors and focus on the aggregation of data coming from different types of sensors in order to enhance the reliability of the results obtained via the video-surveillance.

Author's Publication

Journal Papers

[1] **Antoine Fagette**, Nicolas Courty, Daniel Racoceanu, and Jean-Yves Dufour. Unsupervised dense crowd detection by multiscale texture analysis. *Pattern Recognition Letters*, 2013.

[2] **Antoine Fagette**, Patrick Jamet, Daniel Racoceanu, and Jean-Yves Dufour. Particle video for crowd flow tracking. In *3rd International Conference on Pattern Recognition Applications and Methods*, 2014.

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[4] D. Helbing and P. Molnár. Social force model for pedestrian dynamics. *Physical Review E*, 51:4282, 1995.

[5] P. Sand and S. Teller. Particle video: Long-range motion estimation using point trajectories. *Computer Vision and Pattern Recognition*, 2:2195–2202, 2006.